

Nathan Hutton

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EDUCATION

University of Utah <i>M.S. Computer Science</i> Specialization in Data Visualization & Computer Graphics	Dec. 2025 GPA: 4.0
Westminster University <i>B.S. Computer Science, Applied Mathematics Minor</i> Award: Outstanding computer science student of the year	May 2024 GPA: 4.0

EXPERIENCE

Software Engineer 2 <i>WCF Insurance</i> Program in Java and communicate with business users to solve problems and implement features for in-house insurance software	Jan 2026 – Present <i>Sandy, UT</i>
Software Engineer Intern <i>Flight Safety International</i> Migrated flight simulation's I/O system to Phidgets API in C++, eliminating \$50,000+ in legacy hardware costs	May 2025 – Aug. 2025 <i>Broken Arrow, OK</i>
Software Engineer Intern <i>Idaho National Laboratory</i> - Migrated test harness from MATLAB to Python with Numpy, Scipy, and Pytest, saving thousands in licensing fees - Configured CI/CD GitLab pipelines within a Linux environment to automate build and test processes - Refactored a legacy RF transceiver GUI using Tkinter, improving user workflow and repository maintainability	Feb. 2023 – May 2024 <i>SLC, UT</i>
Computer Science Tutor/TA <i>Westminster University</i>	Aug. 2022 – Dec. 2023 <i>SLC, UT</i>
IT Technician <i>Westminster University</i> - Resolved 500+ Jira tickets for network and account issues - Trained 6 IT employees on Jira ticketing, Windows, and customer service practices	Aug. 2021 – May 2024 <i>SLC, UT</i>

PROJECTS

Real-Time Swarm AI Simulation C++, OpenGL, ImGui Engineered an agent-based AI system from scratch to simulate the emergent behavior of 4,000+ agents	GitHub Video
Solar System C++, OpenGL, JSON Dynamic physics, shadow maps, bloom, and verlet numerical integration	GitHub Video
Volme Renderer C++, OpenGL, Glui Ray marching in GLSL with modifiable transfer functions	GitHub Video
Air Quality Index Visualization Javascript, D3, JSON Interactive D3.js visualization for Utah's AQI, personally contributing 63% of the project's total commits	GitHub Video
Ray/Path Tracer C++ Reflections, refractions, textures, BVHs, anti-aliasing, depth of field, global illumination, and photon mapping	GitHub
Squibblets C#, Unity, Firebase, AGILE Engineered the primary gameplay loop and UI while integrating Firebase for the online leaderboard system, personally accounting for 54% of all commits on a 4-person team	GitHub
Mass Spring System C++, OpenGL, Eigen Simulate a mass spring system made up of 8,000 tetrahedrons in real time	GitHub
Screen-Space Fluid Renderer C++, OpenGL Real-time rendering of cached fluid simulations with caustics. Uses narrow-range filter to achieve surface shape.	GitHub Video

TECHNICAL SKILLS

Languages: C/C++, Java, Python, C#, JavaScript, HTML/CSS, LaTeX, SQL
Tools: Linux, Git, Docker, Vim, VS Code, Visual Studio, Windows, Jira
Libraries: OpenGL, NumPy, Pandas, ImGui, Matplotlib, Eigen, PyTorch, SciPy